Kinetic arrest in a dense active glass: scaling, Koopman prediction, HAVOK and reduced equations

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Abstract. We analyse the kinetic energy K(t) of a two-dimensional Kob–Andersen binary mixture (N = 1000) driven by a persistent active force of magnitude f, using one trajectory for each of 17 forces between 0 and 2. For $1.0 \le f \le 1.65$ the energy decays quasi-exponentially and then collapses abruptly when the system arrests; f = 1.8 is still decaying at the end of the simulated window ($t \approx 3 \cdot 10^4$) and f = 2.0 is nearly stationary. (i) All trajectories with $1.0 \le f \le 1.8$ collapse onto a master curve $\log_{10} K = a(f) + G(t/\tau(f))$ ($R^2 = 0.983$); the arrest time scales with the slow-decay time, measured without using the collapse, as $t_{arr} \propto \tau_K^{0.91}$ ($R^2 = 0.97$), and the final collapse proceeds at a force-independent rate. (ii) The relaxation time grows faster than exponentially (Arrhenius rejected, ΔAIC = 11–26); power-law fits give $f_c \approx 1.8$–1.95 and $\gamma \approx 1.4$–2.1, with a strong f_c–γ degeneracy. (iii) HAVOK reveals strongly non-Gaussian, intermittent forcing in every arresting trajectory (excess kurtosis 3.5–32, far outside phase-randomised surrogates) and Gaussian forcing at f = 2.0; the intermittency appears in the last ~20 % of the slow stage and precedes sudden drops of K at 2–4× chance. (iv) Weak-form sparse regression yields a compact reduced equation $\tau(f)\, dy/dt = g(y - a(f))$, $y = \log_{10} K$, the differential form of the collapse. (v) In free-running, leave-one-force-out prediction, linear Hankel-DMD operators and a rollout-trained deep Koopman autoencoder do not reproduce the arrest, whereas a Koopman model built on the scaling structure predicts the arrest time of unseen forces with a median error of 14% (95% CI 10%–18%). All claims are accompanied by explicit validation tests (Section 5).

1. Introduction

Dense assemblies of self-propelled particles combine glassy caging with persistent driving. Depending on the strength of the active force they either flow as an active fluid or arrest in a mechanically stable, jammed configuration. Existing theoretical descriptions, such as active extensions of mode-coupling theory [8–10] and hydrodynamic theories of active matter [16, 17], do not yet provide a closed prediction for the time dependence of global observables near this arrest. This motivates a data-driven approach.

This report addresses two questions about the mean kinetic energy K(t; f):

- **Q1 – Scaling.** Do trajectories at different active forces obey a common scaling form, i.e. does a master curve exist, and how do the characteristic time scales depend on f?
- **Q2 – Prediction.** Can Koopman-operator methods predict a kinetic-energy time series that was not used to build the model, either from the force alone or from the beginning of the trajectory?

Both questions are approached together with a search for a governing differential equation (sparse identification [3, 4]) and with the Hankel Alternative View of Koopman (HAVOK) [2], which separates linear dynamics from intermittent forcing. Every claim is tested against a baseline, a null model or held-out data; Section 5 collects these tests.

2. System and data

The system is a two-dimensional 65:35 Kob–Andersen Lennard-Jones mixture [1] of N = 1000 particles subject to an active force of magnitude f. For each f ∈ {0, 0.25, 0.5, 0.75, 1.0, 1.05, 1.15, 1.25, 1.35, 1.4, 1.45, 1.5, 1.55, 1.6, 1.65, 1.8, 2.0} one trajectory of the mean kinetic energy K(t) is available, sampled at Δt = 0.5 up to t = 29 505 (59 011 points). All trajectories start from the same value K(0) = 0.1072. We work with $y = \log_{10} K$. After arrest, K decays to a numerical floor of $\sim 10^{-26}$; values below 10^{-20} are discarded. The single final sample (Δt = 0.05) is dropped. For smoothed quantities a centred 41-point moving average is used.

f	regime	t_{arr}	τ_K	τ_{mc}	a(f)
1.00	arrests	1828	215	1432	-0.13
1.05	arrests	2773	398	2563	-0.24
1.15	arrests	3359	381	2160	0.19
1.25	arrests	3331	510	3154	0.36
1.35	arrests	5304	536	3725	0.63
1.40	arrests	6315	654	6315	0.00
1.45	arrests	7582	955	6328	0.54
1.50	arrests	7662	1121	6839	0.76
1.55	arrests	11108	1577	10685	0.62
1.60	arrests	16710	2331	15084	0.70
1.65	arrests	23650	3652	22894	0.76
1.80	slow decay, no arrest by end	none	19226	94461	0.91
2.00	near-stationary	none	311268	–	–

Table 1. Forces analysed in detail (f ≥ 1.0). t_{arr}: time at which K falls below 10^{-20}; τ_K: e-folding time of the slow stage; τ_{mc}, a(f): time scale and offset from the master-curve collapse (reference f = 1.4). Forces f ≤ 0.75 relax like a quench (fast decay with oscillations) and are not part of the scaling analysis.

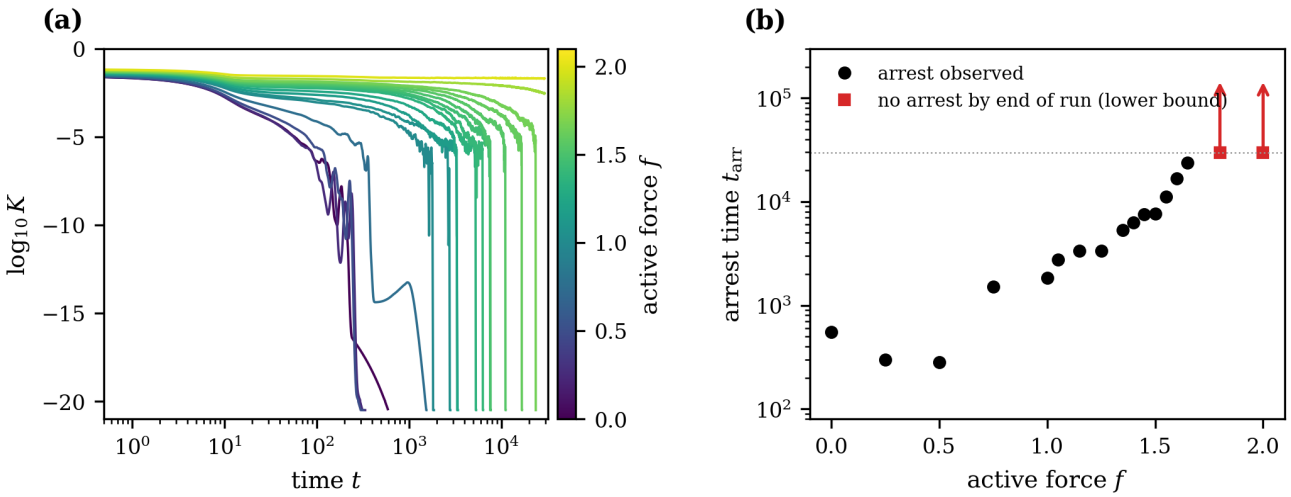


Figure 1. (a) $\log_{10}$K(t) for all 17 forces (smoothed). For f ≥ 1.0 a short transient is followed by a slow, almost exponential decay and an abrupt collapse. (b) Arrest time versus f; f = 1.8 and 2.0 do not arrest within the simulated window (lower bounds).

3. Methods

3.1 Relaxation times and critical fits

Several operational definitions of a relaxation time are used so that conclusions do not depend on one choice: the arrest time t_{arr}; the time at which the smoothed y first reaches −5; the e-folding time τ_K of the slow stage, from a linear fit of ln K between t = 20 and 0.85 t_{arr} (for non-arresting forces, to the end of the run), so that τ_K never uses the collapse itself; and the time scale τ_{mc} from the master-curve collapse. Each is fitted with $\tau = A(f_c - f)^{-\gamma}$, profiling f_c on a grid above the largest force in the fit, and compared with a Vogel–Fulcher-type form $\ln \tau = a + b/(f_c - f)$ and an Arrhenius-type form $\ln \tau = a + bf$ using the Akaike information criterion [14]. Uncertainties come from a non-parametric bootstrap over forces (500–2000 resamples) and from the set of f_c values with ΔAIC < 2.

3.2 Master-curve collapse

Curves are compared on a grid $x = t/\tau(f) \in [0.02, 1]$ for t > 20. The collapse cost is the mean, over x, of the variance of $y - a(f)$ across forces. The time scales τ(f) are optimised (Powell, bounded within a factor e of the initial guess t_{arr}) and, for each set of time scales, the offsets a(f) are obtained by alternating least squares with a(1.4) = 0. Spread values are reported for no rescaling, time rescaling only, and time plus offset.

3.3 HAVOK

Following [2], the fluctuation δy = y − (linear slow trend) is embedded in a Hankel matrix with q = 40 delays (20 time units). The SVD gives coordinates $v_1 \dots v_r$; a linear model $dv_{1..r-1}/dt = A\, v_{1..r-1} + B\, v_r$ is fitted by least squares using fourth-order differences of the SVD time series, and v_r is treated as forcing. The rank r is chosen as the (marginally) stable model with the smallest forced-reconstruction error. Forcing is 'active' when $|v_r| > 2.5\sigma$. All event statistics are causal: a Hankel column is time-stamped at the end of its window, and chance levels are estimated from random times. The implementation was validated on the Lorenz system and against PyDMD (Section 5).

3.4 Equation discovery

Sparse identification [3] is carried out in weak form [4]: $dy/dt = \Theta(y)\xi$ is multiplied by compactly supported test functions $\varphi_k(t) = [(t - a)(b - t)]^4$ (width 100 time units, stride 25) and integrated by parts, so the noisy data are never differentiated. Coefficients are found by sequentially thresholded least squares, with the threshold chosen by BIC using the number of independent windows. Two libraries are used: a generic one, monomials in (y, f) up to degree 2 × 2, and a scaling-informed one, in which $u = y - a(f)$, $s = t/\tau(f)$ and $du/ds = g(u)$ is a polynomial. A stochastic version is obtained from Kramers–Moyal conditional moments [13] of u at a lag of 5 time units.

3.5 Prediction protocols and metrics

All predictions are free-running: after the initial condition, the model receives no data from the trajectory being predicted. In protocol (a), each of the 13 forces with $f \geq 1.0$ is left out in turn, and every fitted quantity (operators, network weights, master curve, $\tau(f)$, $a(f)$, ODE) is rebuilt without it. The first 100 time units are taken as an interpolation of neighbouring training forces. In protocol (b) the first 10, 25 or 50 % of the arrest time of the unseen trajectory is given. Models: parametric Hankel DMD [5, 6, 15] with operators interpolated in f; HDMD fitted on the prefix (equivalent to HAVOK with unknown future forcing set to zero); a force-conditioned Koopman autoencoder [7] trained with a 40-step latent rollout loss; a master-curve Koopman model (one slow mode with eigenvalue $-1/\tau(f)$ from the critical law, followed by the collapse mode); the discovered ODE; and a non-Koopman baseline that copies the nearest training force and stretches its time axis to an interpolated arrest time. Metrics: RMSE, MAE and R^2 of y on the slow stage (up to the earlier of the true and predicted arrest), and the relative error of the arrest time (time at which $y < -10$). A missed arrest counts as 100 % error. Medians over forces are reported with 95 % bootstrap confidence intervals.

4. Results

4.1 Master curve (Q1)

Rescaling time alone reduces the spread of $\log_{10}K$ across the 12 trajectories with $1.0 \leq f \leq 1.8$ from 1.81 to 0.112; adding an offset $a(f)$ reduces it to 0.030, i.e. 98.3 % of the variance is captured by a single function G(x) (Fig. 2a,b). The non-arresting trajectory $f = 1.8$ follows the same G over the range it covers. Because a two-parameter rescaling has considerable freedom, the collapse is tested independently. τ_K is measured from the slow stage only, and the arrest time is then compared with it (Fig. 2c): $t_{arr} \propto \tau_K^{\beta}$ with $\beta = 0.91$ (95 % CI 0.83–0.98, $R^2 = 0.97$). While τ_K varies by a factor 17, the ratio $t_{arr}/\tau_K = 7.8$ varies by only 15% (coefficient of variation). The arrest therefore occurs at an almost fixed point of the slow decay, with a slight sub-linearity ($\beta < 1$). After the cliff, $\log_{10}K$ decreases at 0.40 ± 0.09 decades per time unit for every force ($K \propto e^{-0.91\,t}$, Fig. 2d).

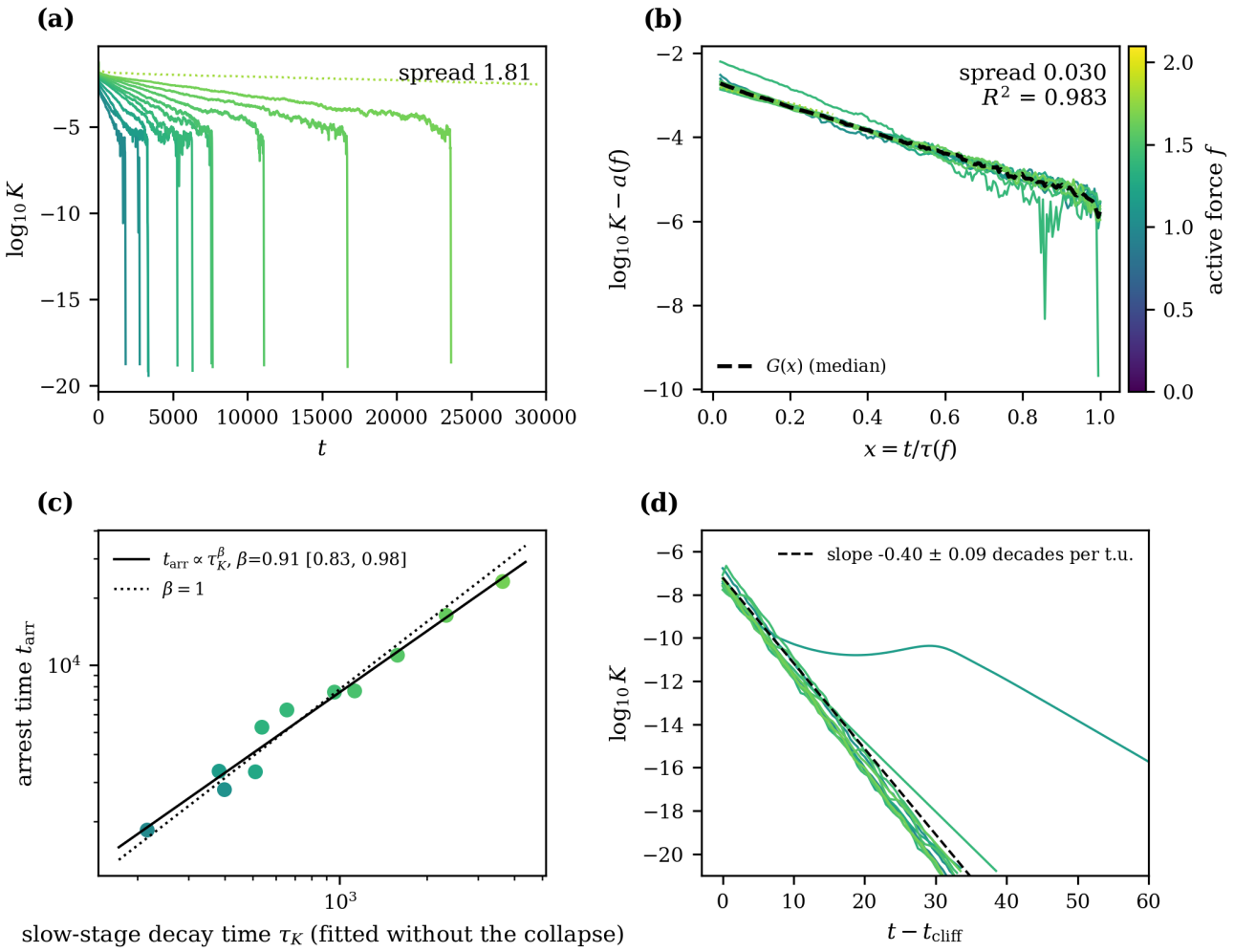


Figure 2. (a) Slow stage of $\log_{10}$K for 1.0 ≤ f ≤ 1.8 (dotted: f = 1.8). (b) Collapse $\log_{10}$K − a(f) versus t/τ(f) with the median master curve G(x). (c) Arrest time versus slow-stage decay time, which is fitted without using the collapse; solid: power-law fit, dotted: proportionality. (d) Final collapse aligned at its onset, with a force-independent slope.

4.2 Critical scaling of the time scale (Q1)

τ definition	n	best f_c	best γ	γ (68 % bootstrap)	f_c, γ with ΔAIC < 2	R^2	ΔAIC Arrhenius	ΔAIC VFT
t_{arr}	11	1.81	1.44	1.16–2.03	1.74–1.94; 1.15–1.97	0.980	11.1	0.8
t(K = 10^{-5})	11	1.85	1.99	1.56–2.67	1.78–2.00; 1.60–2.69	0.986	12.0	0.7
τ_K	12	1.89	1.89	1.22–2.09	1.86–1.93; 1.66–2.18	0.984	24.2	-2.3
τ_{mc}	12	1.93	2.10	1.52–2.41	1.89–2.00; 1.81–2.49	0.983	20.4	-0.9
t(K = 10^{-4})	11	2.02	3.08	2.58–3.79	1.92–2.17; 2.58–3.85	0.996	16.8	1.2
t(K = 10^{-3})	11	3.00	10.66	9.90–11.26	2.53–3.00; 7.57–10.66	0.978	-6.7	5.0
HDMD on K	11	3.00	6.40	6.11–6.62	2.60–3.00; 4.83–6.40	0.990	-6.0	6.6

Table 2. Fits of $\tau = A(f_c - f)^{-\gamma}$. ΔAIC columns are relative to the power law (positive: power law preferred). The first four rows are the robust definitions; the last three (shaded) are biased: times to reach a fixed level mix the f-dependent offset a(f) with the time scale, and HDMD on linear K is dominated by the first few hundred time units.

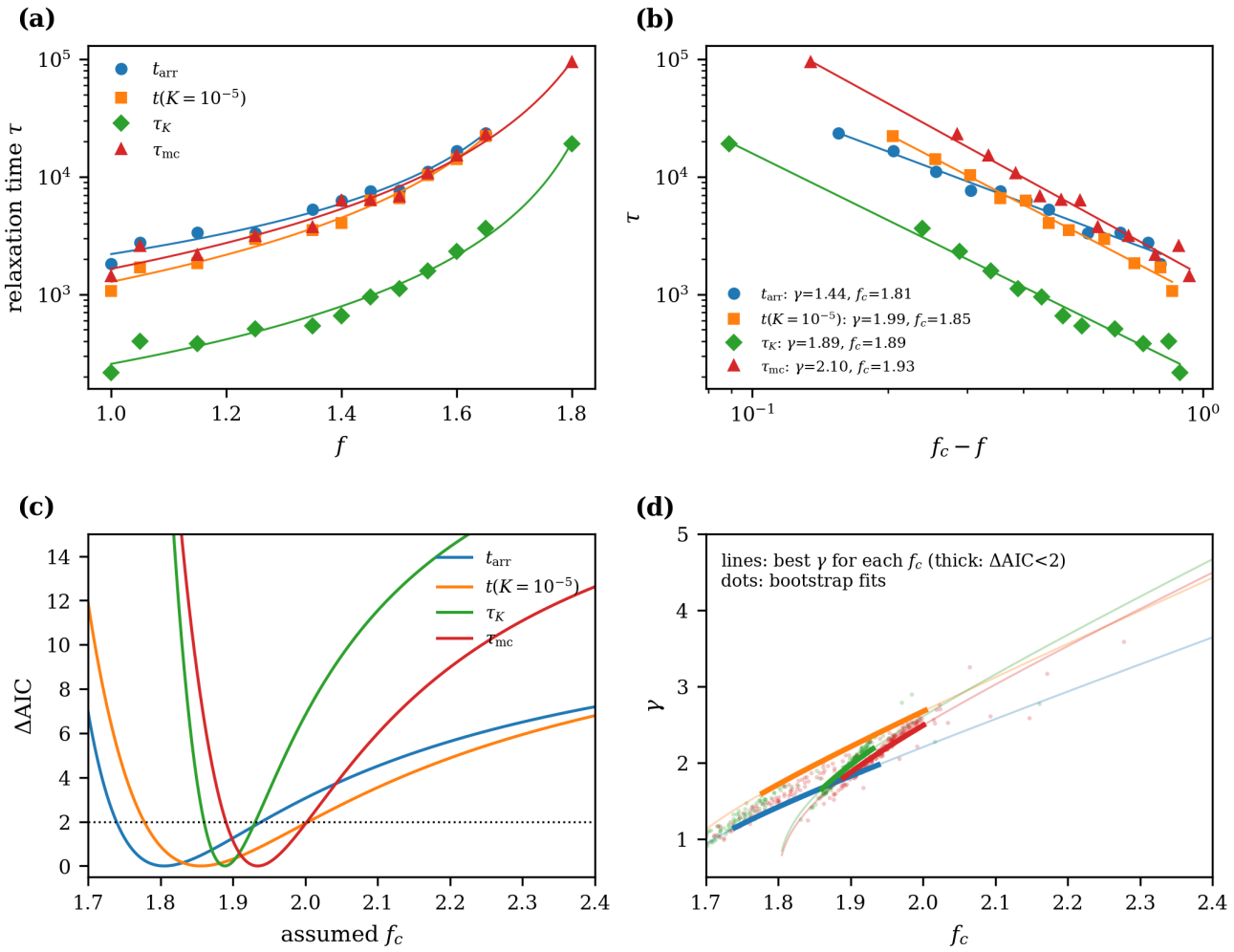


Figure 3. (a) Relaxation times versus f with power-law fits. (b) Same data versus $f_c - f$, each with its own best f_c. (c) ΔAIC of the power-law fit as a function of the assumed f_c. (d) Best γ for each assumed f_c (thick: ΔAIC < 2) and bootstrap fits (dots) for τ_K and τ_{mc}.

For the four robust definitions, an Arrhenius law is rejected (ΔAIC = 11–26), whereas a power law and a VFT-type law describe the data equally well (|ΔAIC| < 2.5). The best estimates span f_c = 1.81–1.94 and γ = 1.44–2.10; Fig. 3d shows that f_c and γ are strongly correlated, so that the data constrain a curve in the (f_c, γ) plane rather than a point. For τ_K, which includes f = 1.8, the result is insensitive to the fitting window (γ = 1.87–1.94, f_c = 1.887–1.893; Fig. S1c). The f = 1.8 trajectory is significantly decaying (slope -2.22×10^{-5} decades per time unit, 95 % CI -2.25×10^{-5} to -2.19×10^{-5}, block bootstrap), which places f_c above 1.8 if this decay ends in arrest (extrapolated $t_{arr} \sim 1.5\times10^{5}$). f = 2.0 also shows a small but statistically significant downward drift (-0.028 decades over the analysed window). If this drift were the same slow decay, f_c would lie above 2.0 (≈ 2.1 with γ ≈ 3); the present data cannot exclude this.

4.3 Intermittency revealed by HAVOK

f	r	forcing kurtosis [95 % CI]	surrogates (max)	dips: preceded / chance	pre-arrest active (local, global baseline)
1.00	4	7.9 [1.2, 11.2]	0.12	55% / 19%	23% (6%, 4%)
1.05	3	6.6 [1.4, 7.7]	0.68	67% / 16%	30% (10%, 4%)
1.15	6	3.5 [0.8, 6.3]	0.21	33% / 27%	22% (7%, 3%)
1.25	5	23.9 [1.8, 42.1]	0.20	17% / 8%	45% (6%, 3%)
1.35	4	6.3 [3.3, 14.6]	0.26	61% / 20%	26% (14%, 5%)
1.40	4	10.7 [5.8, 22.1]	0.31	48% / 16%	14% (18%, 4%)
1.45	4	10.3 [4.1, 16.0]	0.23	36% / 13%	61% (21%, 4%)
1.50	4	26.4 [7.7, 42.7]	0.17	50% / 13%	38% (26%, 4%)
1.55	13	29.3 [3.4, 36.6]	0.12	37% / 14%	68% (19%, 3%)
1.60	7	13.9 [7.0, 19.7]	0.11	68% / 16%	41% (33%, 4%)
1.65	3	31.6 [12.8, 49.3]	0.09	24% / 9%	56% (49%, 3%)
1.80	4	0.6 [0.4, 0.8]	0.09	0% / 16%	–
2.00	4	-0.1 [-0.2, 0.0]	0.11	0% / 10%	–

Table 3. HAVOK diagnostics (q = 40). Kurtosis: excess kurtosis of the forcing v_r; 0 for a Gaussian. Surrogates: largest value among 19 phase-randomised surrogates with the same power spectrum. Dips: fraction of strong downward excursions of K preceded within 20 time units by active forcing, versus the same fraction at random times. Pre-arrest: fraction of time with active forcing 120–20 time units before the collapse, compared with the preceding 500 time units (local) and the whole trajectory (global).

The forcing is strongly non-Gaussian for every arresting force and lies far outside the surrogate distribution (all 19 surrogates have |kurtosis| < 0.7, Fig. 4c). It is weakly non-Gaussian at f = 1.8 (0.58, 95 % CI 0.39–0.78) and Gaussian at f = 2.0, where it is indistinguishable from the surrogates. Active forcing is essentially absent in the first 60 % of the slow stage and appears in its last ~20 % ($x = t/\tau \geq 0.8$, $K \sim 10^{-5}$), together with growing fluctuations of log K (Fig. 4a). Sudden drops of K are preceded by active forcing 2.9× more often than random times (median over forces). In the 100 time units before the collapse, active forcing is 10× more frequent than on average over the trajectory, but only 2.9× more frequent than in the preceding 500 time units (medians). The pre-arrest enhancement thus largely reflects the general rise of intermittency as K approaches the arrest level; a specific burst immediately before arrest is clear only for $f \leq 1.45$. These conclusions are unchanged for q = 20 and 80 (Table 9). The linear part of the HAVOK model is marginally stable for most forces and is used here as a diagnostic rather than as a forecaster.

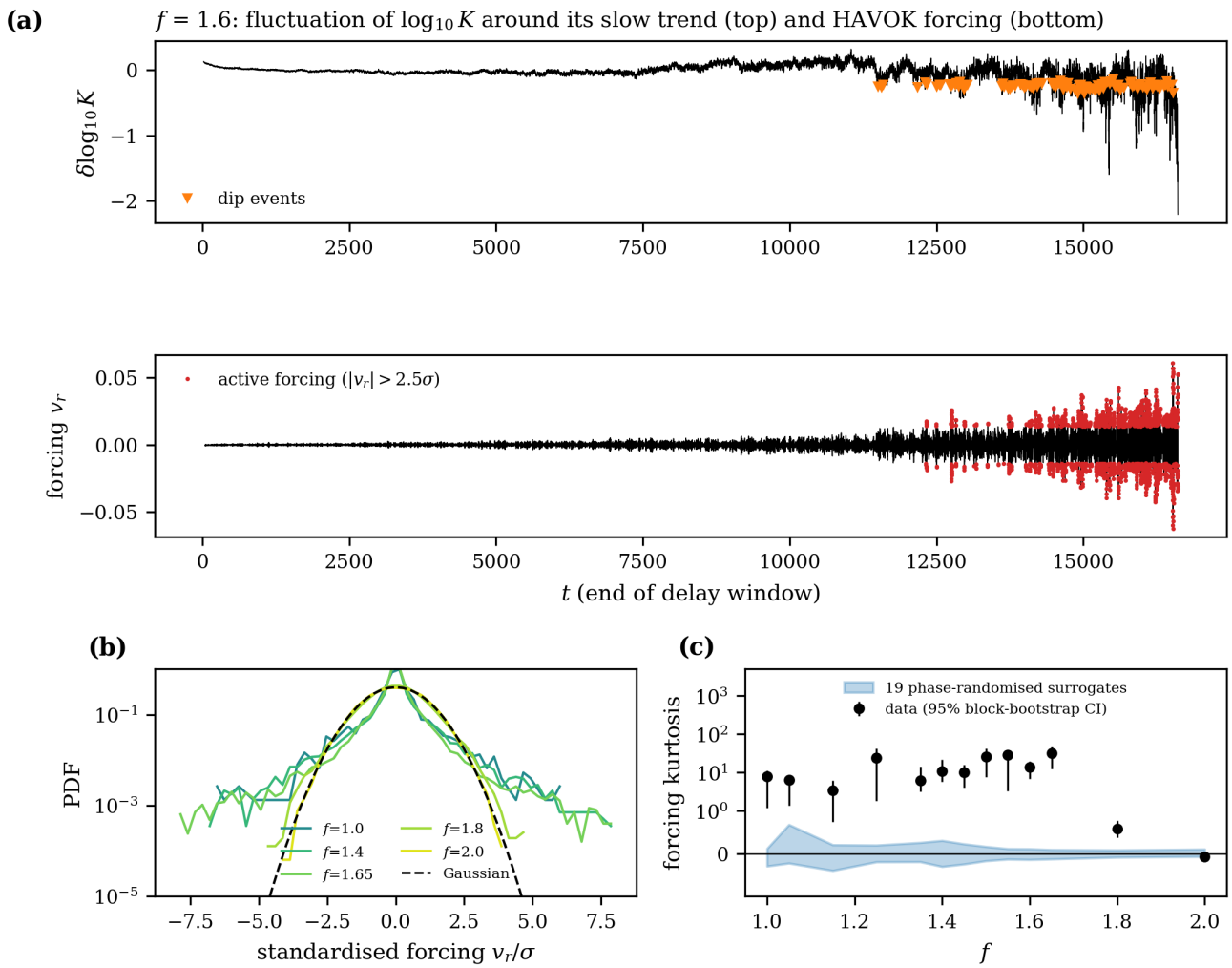


Figure 4. (a) f = 1.6: fluctuation of $\log_{10}$K around its slow trend with detected dips, and the HAVOK forcing v_r (red: active). (b) Distribution of the standardised forcing for several forces, with a Gaussian for reference. (c) Excess kurtosis of the forcing with 95 % block-bootstrap intervals; the band spans 19 phase-randomised surrogates per force.

4.4 Reduced governing equation

With the generic library in (y, f), weak SINDy and a strong-form PySINDy fit return dense equations whose coefficients disagree with each other and which diverge when integrated for f ≥ 1.55. A polynomial in f cannot represent a rate that diverges at f_c. With the scaling-informed library, the selected model is

$$\tau(f)\, dy/dt = g(y - a(f)),\ g(u) = 7.10\, u + 2.86\, u^2 + 0.307\, u^3,$$

with $\tau(f) = 1433\ (1.935 - f)^{-2.10}$, a(f) = 1.36 f -1.47, followed by dy/dt = −0.40 once u < -6.9. This equation is the differential form of the collapse of Section 4.1, written with about eight constants. It is a reduced empirical model for one global observable, not a microscopic equation of motion.

term	coefficient	95 % bootstrap CI	selected in
1	0.000	[-12.02, 64.80]	18%
u	7.102	[-4.83, 50.00]	99%
u^2	2.859	[-0.62, 11.73]	98%
u^3	0.307	[0.00, 0.89]	89%

Table 4. Coefficients of g(u) with bootstrap over forces (200 resamples). The cubic structure is selected in 89–99 % of resamples; individual coefficients are correlated and therefore broad, while g(u) itself is well constrained in the data range (Fig. 5a).

Validated leave-one-force-out, with the equation re-derived each time, the model reproduces $\log_{10}K$ on the slow stage with a median RMSE of 0.38 decades and the arrest time with a median error of 17% (Fig. 5b). A quadratic g is clearly worse (37% versus 17% for the cubic and 14% for a quartic, in a common leave-one-out test). |g| has a minimum near $u \approx -4.6$, a bottleneck that coincides with the onset of intermittency (Section 4.3). The hypothesis $dK/dt = -AK + B\,f\,K^n$ is not supported: n is not identifiable (residual changes by < 15 % for n = 0–3), the best fit has $B = -8.3\times10^{-6} < 0$, and arrest times are off by 50–200 %.

The Kramers–Moyal analysis shows that the diffusion coefficient of u grows by about three orders of magnitude as K approaches the arrest level (Fig. 5c). The resulting Langevin model $du = g(u)\,ds + (2D_2(u))^{1/2}\,dW$ reproduces the median rescaled arrest time ($s^* = 1.21$ versus 1.17 observed), but its 16–84 % range (0.66–2.11) is wider than the observed spread (1.04–1.27). The diffusion estimate is inflated by the intermittent jumps, which a Gaussian Langevin model cannot represent (Fig. 5d).

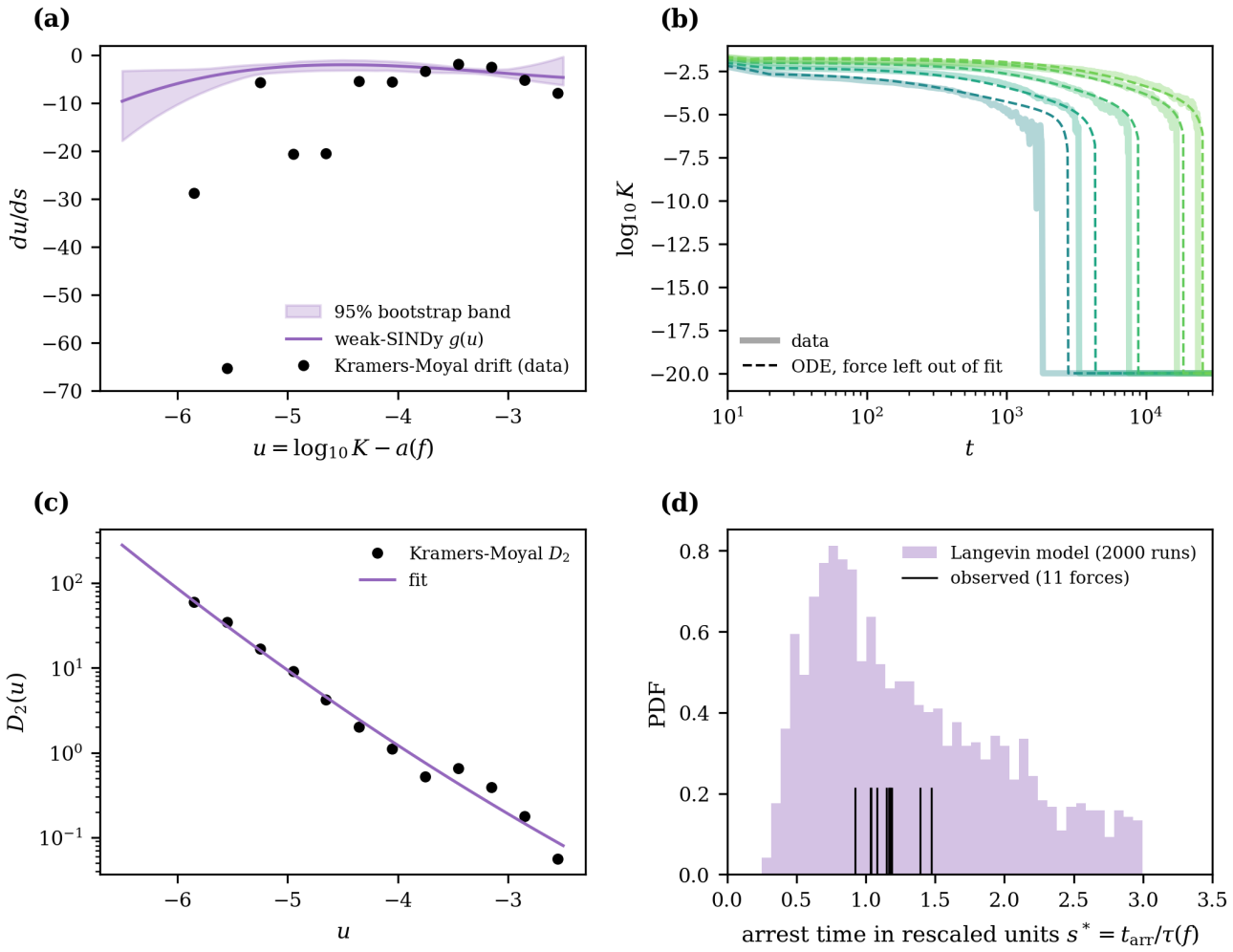


Figure 5. (a) g(u) from weak SINDy with a 95 % bootstrap band, and Kramers–Moyal drift estimates (which are biased low near arrest by jump events). (b) Leave-one-force-out integration of the equation (dashed) against the data. (c) Kramers–Moyal diffusion coefficient. (d) Rescaled arrest times: Langevin model versus the 11 observed values.

4.5 Prediction of unseen trajectories (Q2)

model	median $\lvert\Delta t_{arr}\rvert/t_{arr}$ [95 % CI]	within 25 %	missed arrests	false arrests	RMSE (dec.)	MAE (dec.)	R^2
Neighbour baseline	8% [3%, 24%]	10/11	0/11	1/2	0.36	0.16	0.89
Parametric HDMD	67% [59%, 73%]	0/11	1/11	0/2	4.53	4.20	< 0
Deep Koopman AE	100% [100%, 100%]	0/11	11/11	0/2	3.86	3.58	< 0
Master-curve Koopman	14% [10%, 18%]	10/11	0/11	0/2	0.31	0.17	0.87
Discovered ODE	17% [9%, 31%]	7/11	0/11	0/2	0.45	0.26	0.80

Table 5. Protocol (a): prediction of a whole trajectory from the force alone, leave-one-force-out over 13 forces (11 arresting, 2 non-arresting). RMSE, MAE and R^2 refer to $\log_{10}K$ on the slow stage of the arresting forces (medians). A missed arrest counts as 100 % error.

f	Neighbour baseline	Parametric HDMD	Deep Koopman AE	Master-curve Koopman	Discovered ODE
1.0	+47%	-48%	no arrest	+10%	+67%
1.05	-24%	-74%	no arrest	-41%	-22%
1.15	-11%	-73%	no arrest	-24%	-5%
1.25	+25%	-59%	no arrest	-2%	+31%
1.35	-4%	-67%	no arrest	-14%	+11%
1.4	+0%	-67%	no arrest	-14%	+9%
1.45	-8%	-69%	no arrest	-12%	+17%
1.5	+20%	-57%	no arrest	+11%	+45%
1.55	+2%	-67%	no arrest	-4%	+31%
1.6	-3%	-63%	no arrest	-17%	+11%
1.65	+6%	no arrest	no arrest	-18%	+8%
1.8	false arrest	correct (none)	correct (none)	correct (none)	correct (none)
2.0	correct (none)	correct (none)	correct (none)	correct (none)	correct (none)

Table 6a. Relative arrest-time error per left-out force (protocol a). f = 1.0 and f ≥ 1.65 require extrapolation in f.

Linear Koopman operators (parametric HDMD) decay smoothly and never produce the abrupt arrest, so they predict it far too early or not at all. The rollout-trained Koopman autoencoder drifts in free rollout and misses all 11 arrests. The master-curve Koopman model and the discovered equation predict all arrests, with median errors of 14% and 17%, and no false arrest for f = 1.8 or 2.0. The neighbour baseline attains a lower median arrest-time error (8%), because t_{arr} varies smoothly with f between training forces. It is, however, itself a one-parameter collapse (a neighbouring trajectory stretched in time), so its success is further evidence for the common shape of the trajectories. It differs from the master-curve model in how τ(f) is supplied: by local interpolation rather than by the critical law. The critical law matters where interpolation is not available: at f = 1.0 the baseline errs by 47% against 10%, and at f = 1.8 the baseline predicts a spurious arrest. The master-curve model also reproduces the slow stage more closely (RMSE 0.31 versus 0.36 decades). The bootstrap intervals of the two arrest-time medians overlap; with 11 arresting forces neither model can be declared more accurate for interior forces.

model	10 % given	25 % given	50 % given
HDMD on prefix (≈ HAVOK, zero forcing)	100% [48%, 100%] (6 missed)	100% [57%, 100%] (6 missed)	100% [38%, 100%] (3 missed)
Deep Koopman AE	100% [100%, 100%] (11 missed)	100% [100%, 100%] (10 missed)	41% [25%, 100%] (4 missed)
Master-curve Koopman	20% [13%, 31%] (0 missed)	15% [4%, 25%] (0 missed)	13% [8%, 22%] (0 missed)
Discovered ODE	60% [41%, 64%] (1 missed)	53% [33%, 85%] (1 missed)	31% [11%, 40%] (1 missed)

Table 6b. Protocol (b): median relative arrest-time error [95 % CI] when the first part of the unseen trajectory is given.

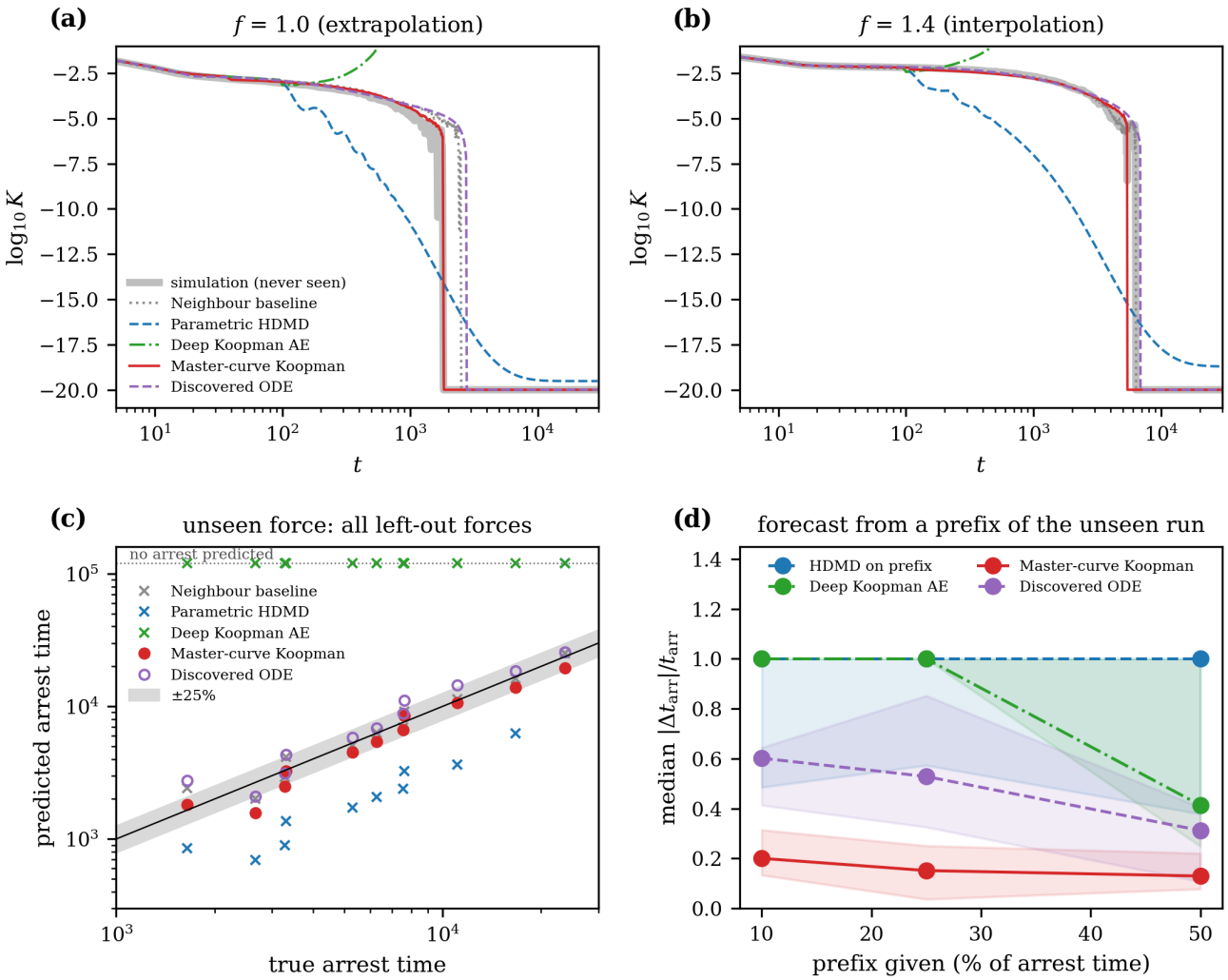


Figure 6. Free-running prediction of trajectories never used in training. (a) f = 1.0, which requires extrapolation in f, and (b) f = 1.4. (c) Predicted versus true arrest time for all left-out forces (crosses at the top: no arrest predicted). (d) Median arrest-time error with 95 % bootstrap bands as a function of the given prefix.

4.6 One-step-ahead versus free-running evaluation

A sequence model can be evaluated by predicting one step ahead from the observed history at every step, or by running it freely from an initial window. The two can differ by orders of magnitude, so the protocol must match the question asked. For the force-conditioned autoencoder trained with a one-step loss (data/processed/koopman_model.pth), the one-step-ahead error on the original held-out forces is 0.02–0.14 decades, whereas the free-running error over 10^4 time units is 3.5–11.7 decades (Table 7, Fig. 7). All prediction results in Section 4.5 are therefore free-running.

f	one-step-ahead RMSE	free-running RMSE
0.5	0.141	8.77
1.25	0.076	3.50
1.6	0.019	5.73
1.8	0.020	11.75

Table 7. RMSE of $\log_{10}(K + 10^{-12})$ over the first 10^4 time units for the one-step-trained autoencoder.

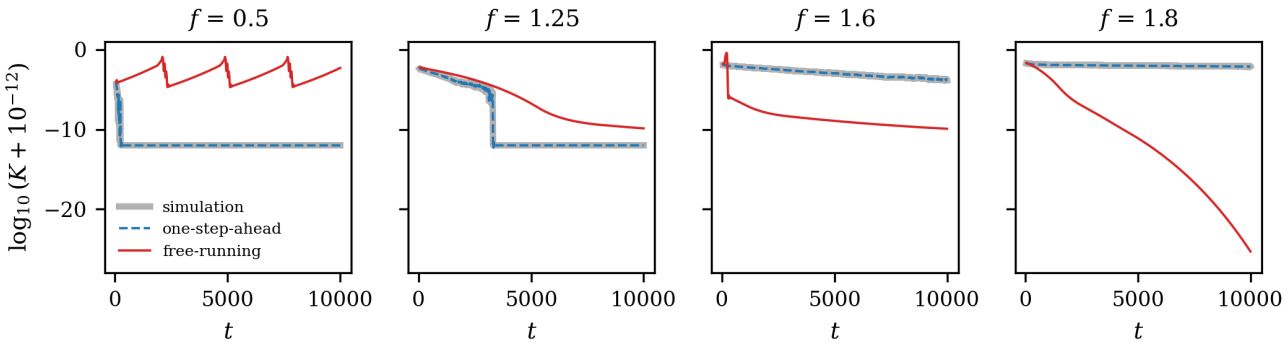


Figure 7. The same one-step-trained autoencoder evaluated one step ahead (dashed) and free-running from the first window (red).

5. Validation of the claims

Each claim is listed with the test used to check it, the outcome, and a verdict. 'Supported' means the test could have failed and did not; 'Partly supported' means the claim holds only in a qualified form.

#	claim	test	result	verdict
1	f = 1.65 arrests; f = 1.8 is still in the slow decay at the end of the run	Arrest criterion $K < 10^{-20}$; OLS slope of log K for t ≥ 5000 with moving-block bootstrap	$t_{arr}(1.65) = 23650$; slope(1.8) = -2.22×10^{-5} [-2.25×10^{-5}, -2.19×10^{-5}], p < 0.001	Supported
2	$f_c > 1.8$	Same as 1, plus extrapolation of the slow decay	Requires the 1.8 decay to end in arrest (expected t ~ $1.5\cdot10^5$, not simulated); f = 2.0 also drifts slightly (-0.028 decades)	Partly supported
3	τ diverges faster than exponentially	AIC: power law versus Arrhenius, 4 τ definitions	ΔAIC = 11–26 in favour of a divergent form	Supported
4	$f_c \approx 1.8$–1.95, $\gamma \approx$ 1.4–2.1	Profile over f_c, bootstrap over forces, fit-window variation	Robust to window; f_c–γ degenerate; power law ≈ VFT	Partly supported (range only)
5	Master curve for 1.0 ≤ f ≤ 1.8	Collapse R^2; arrest predicted from slow-stage τ_K alone; held-out prediction (Table 5)	$R^2 = 0.983$; β = 0.91 [0.83, 0.98]; CV(t_{arr}/τ_K) = 15%	Supported (β slightly < 1)
6	Force-independent final collapse rate	Slope of log K after the cliff, 11 forces	0.40 ± 0.09 decades per time unit	Supported
7	Intermittent (non-Gaussian) forcing in arresting trajectories, Gaussian at f = 2.0	19 phase-randomised surrogates per force; block-bootstrap CI; q = 20/40/80	Data above all surrogates for all 11 arresting forces and f = 1.8; f = 2.0 consistent with surrogates	Supported
8	Forcing precedes drops of K	Causal hit rate versus random-time chance level	2.9× chance (median), stable over q	Supported
9	A specific burst just before arrest	Pre-arrest activity versus the preceding 500 time units	2.9× (median); clear only for f ≤ 1.45	Partly supported
10	Reduced equation τ(f) dy/dt = g(y − a(f)) predicts unseen forces	Leave-one-force-out integration; bootstrap of coefficients; polynomial degree; synthetic recovery	Median arrest error 17%; structure kept in 89–99 % of bootstraps; recovery exact up to 2× the data noise	Supported as empirical model
11	$dK/dt = -AK + BfK^n$	Scan of n, sign of B, arrest-time error	n not identifiable, B < 0, errors 50–200 %	Not supported
12	Langevin model reproduces arrest statistics	Simulated versus observed rescaled arrest times	Median matches; spread overestimated	Partly supported
13	Plain Koopman operators do not predict the arrest	Free-running leave-one-out (Table 5)	Parametric HDMD 67%, autoencoder 100% median error	Supported
14	Scaling-based Koopman model predicts unseen forces	Free-running leave-one-out; bootstrap CI; comparison with a neighbour baseline	14% [10%, 18%], no missed or false arrests; time-stretched neighbour baseline 8% but one false arrest and larger errors under extrapolation	Supported; no advantage over interpolation for interior forces
15	Method implementations are correct	HAVOK on Lorenz; comparison with PyDMD; weak-SINDy recovery	Lorenz forcing precedes switches at 2.7× chance; PyDMD eigenvalue agrees to < 1 %; correct terms recovered at 0.05–0.2 decades noise	Supported

Table 8. Claims, tests and verdicts.

delay window q	median kurtosis (arresting)	kurtosis f = 1.8	kurtosis f = 2.0	dips: hit/chance	pre-arrest / local
20	11.7	0.71	-0.00	3.3	2.0
40	10.7	0.58	-0.07	2.9	2.9
80	9.8	0.46	-0.03	3.1	2.6

Table 9. Sensitivity of the HAVOK results to the delay window.

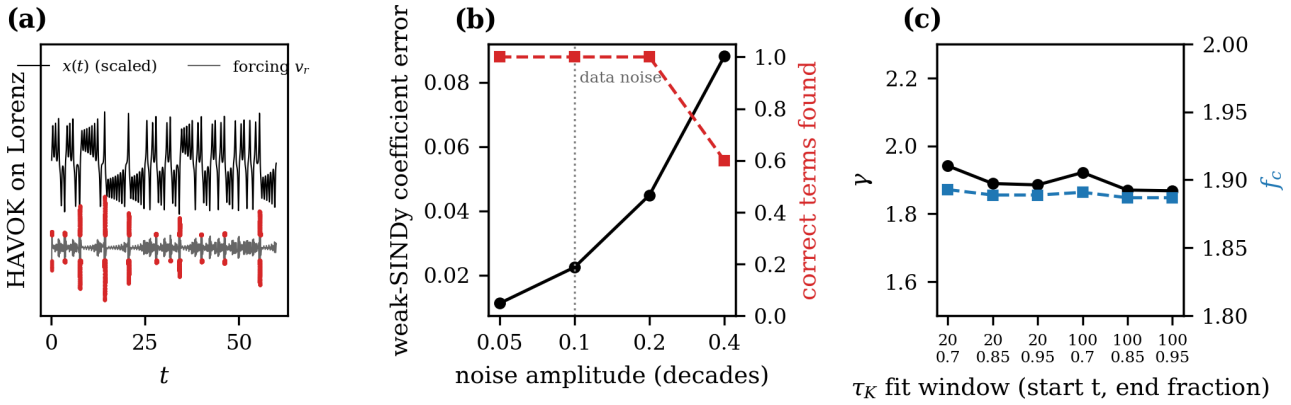


Figure S1. Method validation. (a) HAVOK on the Lorenz system: the forcing (grey, active in red) is concentrated at lobe switches of x(t). (b) Weak SINDy on synthetic data with the same correlated noise as the simulations: coefficient error and fraction of correctly identified terms versus noise amplitude (the data correspond to ≈ 0.1 decades). (c) γ and f_c from τ_K for six choices of the slow-stage window.

6. Discussion

The slow stage of the arresting trajectories is an almost exponential decay of K whose rate, $1/\tau_K$ = 1/215 … 1/19226, is two to four orders of magnitude smaller than the post-arrest relaxation rate (0.91). In an energy balance $dK/dt = -\Gamma K + \Pi_{act}$, the active power input therefore compensates all but a fraction $1/(\Gamma\tau_K)$ of the dissipation during the slow stage and vanishes once the system is trapped. As f approaches f_c this compensation becomes complete, which is the origin of the diverging τ. The onset of intermittency at a fixed point of the rescaled decay, together with the bottleneck in g(u), suggests that arrest is preceded by a growing population of transient, partially arrested configurations.

These results are phenomenological. They characterise one global observable in one model system and do not replace a microscopic theory. They do provide quantitative targets for such theories: a diverging time scale with γ ≈ 1.4–2.1 (to be compared with mode-coupling-type predictions [8–11]), a common shape of the decay, a force-independent collapse rate, and intermittency that grows as arrest is approached.

Limitations

- One trajectory per force; run-to-run scatter (the Langevin model suggests ±25 % in t_{arr}) cannot be measured.
- One density, system size and persistence time, apparently from a single initial configuration; universality across systems is not tested.
- f_c and γ are degenerate, and a power law cannot be distinguished from a VFT law; whether f = 1.8 and 2.0 eventually arrest is not known.
- All validation is within this dataset (leave-one-force-out); no blind prediction of new simulations has been made.
- The quench-like regime f ≤ 0.75 is not covered by the scaling description.

7. Conclusions

Q1 – Scaling. Yes, within this dataset. For $1.0 \le f \le 1.8$ the kinetic energy follows $\log_{10} K = a(f) + G(t/\tau(f))$ with a common shape G, an arrest at an almost fixed rescaled time, a force-independent final collapse, and a time scale that diverges faster than exponentially ($f_c \approx 1.8$–1.95, $\gamma \approx 1.4$–2.1, degenerate).

Q2 – Koopman prediction. Linear Hankel-DMD operators and a deep Koopman autoencoder, run freely, do not predict unseen trajectories because they cannot generate the arrest. A Koopman model built on the scaling structure does: it predicts the arrest time of left-out forces with a median error of 14%, and of partially observed trajectories with 20% (10 % given) to 13% (50 % given). For forces inside the training range, a time-stretched copy of the nearest training trajectory, which is itself a one-parameter collapse, is at least as accurate (8%); the critical law adds predictive value under extrapolation in f.

Equation and HAVOK. The dynamics of log K are captured by the reduced equation $\tau(f)\, dy/dt = g(y - a(f))$, with a cubic g. HAVOK identifies intermittent forcing that is specific to the arresting regime and switches on in the last part of the slow decay.

8. Recommended next steps

- Several independent runs per force, to attach error bars to t_{arr}, τ, f_c and γ.
- A blind test: predict t_{arr} for f = 1.7, 1.75, 1.85 and 1.9 with the reduced equation before simulating them, and run these forces for longer than 10^5 time units.
- A second system size and a second density or persistence time, to test whether G and the collapse rate are system-independent.
- Comparison of γ and the shape of G with active mode-coupling and random first-order transition predictions.

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Appendix A. Reproducibility

All results are produced by the scripts in src/, run from the project root in this order: scaling_analysis.py, havok.py, equation_discovery.py, koopman_forecast.py (about 30 minutes on an NVIDIA RTX 3060), validation.py, report_figures.py. Numerical results are stored in data/processed/*.json and are read directly by reports/build_technical_report.py, which generates this document. Main settings: smoothing window 41 samples; slow stage $t \in [20, 0.85\, t_{arr}]$; HAVOK $q = 40$, rank 3–15 selected by stability and reconstruction, active threshold 2.5σ, look-back 20 time units; weak SINDy test-function width 200 samples, stride 50, BIC on n/4 independent rows; forecasting grid $\Delta t = 5$, initial window 100 time units; autoencoder latent dimension 10, hidden width 64, 30 epochs, 40-step rollout loss; bootstrap resamples 500–5000. Software: Python 3.13, NumPy, SciPy, PyTorch 2.7, PyDMD 2025.8 and PySINDy 2.1 (cross-checks).